

ISAS Challenge 2026: Directional Long-Context Specialist Correction for Imbalanced BLE Location Prediction

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Abstract. Bluetooth Low Energy (BLE) location prediction is difficult under sparse observations, unstable received signal strength, overlapping windows, and severe class imbalance. We propose Directional Long-Context Specialist Correction (DLCS), a dual-timescale system that keeps a 22-class hierarchical Random Forest as the default predictor and permits a 14-room, 50-s specialist to modify only selected uncertain decisions. The main path uses 175 features from midpoint-labeled 10-s windows; the specialist uses 119 label-free context features. Repeated session-grouped validation increases Macro-F1 from 0.5367 ± 0.0116 to 0.5487 ± 0.0158 , but a visit audit finds that 24.1% of labeled visits are split across session groups, exposing 44.3–44.8% of validation windows to another window from the same visit in training. We therefore add fixed-22-class leave-one-day-out (LODO) evaluation. DLCS improves raw LODO Macro-F1 from 0.3647 to 0.3680, whereas after common temporal smoothing the main model scores 0.4026 and DLCS 0.3993. The results support selective long-context correction within a calibrated deployment period, but do not establish a cross-day gain for the final smoothed system.

Keywords: BLE indoor localization; RSSI fingerprinting; class imbalance; temporal context; specialist model; selective correction.

1. Introduction

BLE localization is suitable for care facilities because it is low-cost and avoids privacy-sensitive video, but RSSI varies with occlusion, multipath propagation, body orientation, crowding, and packet loss [4, 5]. These effects are well documented in broader indoor-positioning surveys [11]. The ISAS Challenge 2026 dataset also has a strong long-tail distribution: common areas contain thousands of fingerprints, whereas several patient rooms contain only a few dozen. Earlier work in the same nursing-facility BLE setting investigated signal-pattern-guided relabeling [1]. Severe class imbalance is a well-established learning challenge [2], and ensemble-based remedies have been studied extensively [3].

This paper addresses only the challenge location-prediction task. A single global classifier must separate high-support common areas and low-support rooms, while a long-context model can stabilize transient signal loss but may mix transitions or change already reliable predictions. The proposed novelty is therefore directional specialization: (i) a reproducible 10-s main representation preserves brief visits; (ii) a separate label-free 50-s representation supplies patient-room context; and (iii) a frozen, interpretable gate allows corrections only for validated uncertain confusion patterns. We also audit the evaluation unit directly and distinguish within-period performance from cross-day robustness.

2. Signal Construction

The source archive contains timestamped BLE events and manually recorded location intervals. Only the 25 configured beacon MAC addresses are retained. The main dataset uses 10-s windows with a 2-s stride. For a window $[t, t + 10)$, the target is the first interval containing the midpoint $t + 5$; windows without a valid midpoint label are discarded. Consecutive windows overlap by 8 s, preserving short visits but creating dependence that must be handled during validation.

For each beacon, the current window contributes mean RSSI, presence, population standard deviation, maximum RSSI, and packet count. Two history blocks summarize the previous eight retained windows in the same contiguous sequence: past mean packet count and past mean RSSI standard deviation. Missing mean and maximum RSSI values are encoded as -125 ; the remaining missing features are zero. The final main vector has 175 numeric features. The dataset contains 15,382 fingerprints from 22 classes, with support ranging from 15 to 5,375.

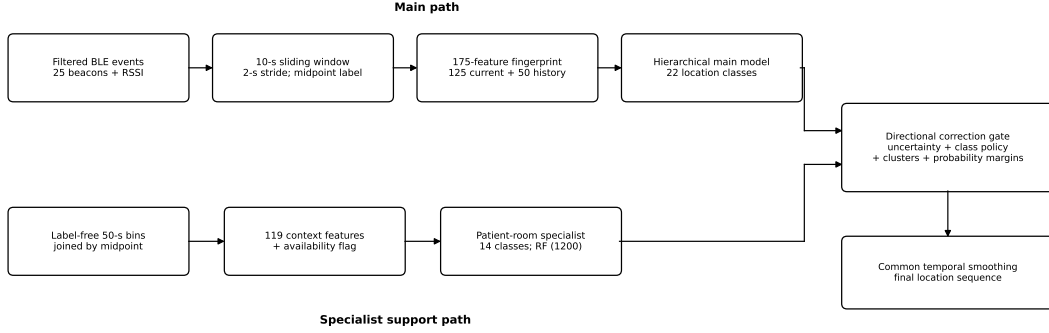
The specialist uses non-overlapping, label-free 50-s bins. Each bin contains four statistics for each beacon (mean, count, maximum, and standard deviation) plus 19 global and cyclical time descriptors, giving 119 context features and an availability flag. Each 10-s fingerprint is joined to the 50-s bin containing its midpoint; the supervised target remains the 10-s midpoint label, so no majority label is assigned to the longer context.

3. Directional Long-Context Specialist Correction

Figure 1 summarizes the complete architecture.

Table 1. Main-path feature blocks.

Feature block	Count	Definition
Current mean RSSI	25	Mean signal strength in the 10-s window
Presence	25	Beacon-observed indicator
Current RSSI standard deviation	25	Within-window population dispersion
Current maximum RSSI	25	Strongest signal in the window
Current packet count	25	Number of received events
Past mean packet count	25	Mean over the previous eight retained windows
Past mean RSSI standard deviation	25	Mean dispersion over the previous eight windows
Total	175	Metadata and labels are stored separately

**Figure 1.** DLCS pipeline. The 22-class main path retains ownership of the prediction; the 50-s patient-room specialist contributes only when the directional gate approves a correction.

3.1. Hierarchical main model

The main classifier is a two-level Random Forest. Tier 1 separates numbered patient rooms from common areas, and Tier 2 predicts the class within the routed group. Each forest uses 400 trees, square-root feature subsampling, and balanced-subsample class weights. Borderline-SMOTE is fitted only inside each training fold and raises classes below 200 training samples to a target of 200. Imputation, resampling, and model fitting are isolated from validation data [6, 7, 8].

3.2. Patient-room specialist and gate

The specialist is a 1,200-tree Random Forest over 14 rooms: 501, 502, 503, 506, 508, 510, 511, 512, 513, 515, 518, 520, 522, and 523. It is not a global replacement because a 50-s bin can span a transition. Let $p_m(c | x)$ be the main probability normalized over specialist classes and $p_s(c | z)$ the specialist probability. Candidate scores are

$$q(c) = 0.25 p_m(c | x) + 0.75 p_s(c | z). \quad (1)$$

The main decision changes only when context is available, the main top two classes are specialist rooms, the main margin is at most 0.30, the source is not protected, the target is validated, the source–target pair belongs to a frozen confusion cluster, and the candidate exceeds the source by at least 0.03 (with stricter pair-specific margins where required). Protected sources are 501, 502, 516, 522, and 523; validated targets are 508, 510, 513, 515, 518, and 520. The policy was selected on development repetitions and frozen before confirmation. This selective, asymmetric intervention is the central difference from global blending or a conventional mixture-of-experts router [10].

3.3. Temporal smoothing

Both main and DLCS predictions receive the same centered majority filter, so their paired difference isolates the gate. The default radius is 15 fingerprints, hallway uses radius 2, and short segments of rooms 501, 513, and 515 are retained when local confidence is insufficient. Because future samples are used, this stage is offline and non-causal.

4. Evaluation Protocol

All metrics use labeled training data; the organizer-held test set is excluded from model selection and reporting. First, repeated stratified group five-fold validation with seeds 10–13 reproduces the development protocol. Groups are contiguous runs of retained windows, and all specialist training rows sharing an exact 50-s bin with validation are purged. Main and DLCS are evaluated on identical out-of-fold rows.

A direct audit then maps each midpoint back to its labeled visit. Of 332 represented visits, 80 (24.1%) are split into multiple session groups. Across the four repeated splits, 44.3–44.8% of validation windows have another window from the same visit in training. Exact 50-s-bin purging therefore removes one leakage channel but not same-visit exposure between adjacent bins. We consequently use four-fold LODO as the cross-day robustness check: each date is held out once, the other three dates are used for fitting, and Macro-F1 is computed over the fixed 22-class

set. This follows the principle that validation groups should match statistically independent deployment units [9]. The gate and window configuration were frozen from earlier development, so LODO is a robustness audit rather than fully nested day-causal model selection.

The fingerprint builder was rerun from the raw events and labels and reproduced the distributed table’s SHA-256 digest. An interval audit found 14,032 windows (91.2%) fully contained in their assigned interval, 598 (3.9%) intersecting multiple labeled locations, and 752 (4.9%) only partially overlapping one labeled interval. These windows are retained to reproduce the evaluated pipeline but represent transition or boundary noise.

Table 2. Construction and leakage audit.

Audit item	Result	Audit item	Result
Candidate / retained windows	47,720 / 15,382	Classes / features	22 / 175
Smallest / largest support	15 / 5,375	Represented visits	332
Visits split across groups	80 (24.1%)	Same-visit exposure	44.3–44.8%
Shared exact 50-s bins	0 in every fold	LODO folds	4

5. Results and Discussion

Table 3 contains the principal results. Under the repeated session-grouped diagnostic, the hierarchical model outperforms flat tree baselines and DLCS adds a paired Macro-F1 gain of 0.0120. Because same-visit exposure is substantial, the absolute session scores should be read as within-period diagnostics. Under LODO, the gate gives a small raw gain of 0.0033 Macro-F1. Common smoothing has a larger effect, raising the main model to 0.4026, but the corrected sequence reaches 0.3993. Thus the new specialist mechanism is useful before smoothing across days, but the final system does not preserve that gain. The smoothed main/DLCS Macro-F1 pairs for 10–13 April are 0.2337/0.2337, 0.3644/0.3644, 0.3791/0.3610, and 0.3558/0.3707, showing that the gate helps the final day, harms 12 April, and is neutral on the first two days after smoothing.

Table 3. Controlled session-grouped diagnostics and fixed-22-class LODO robustness results.

System	Macro-F1	Accuracy	Balanced acc.
<i>Repeated session-grouped diagnostic (mean $\pm$ SD)</i>			
Balanced linear	0.4489 $\pm$ 0.0335	0.6809 $\pm$ 0.0096	–
Balanced Extra Trees	0.4629 $\pm$ 0.0142	0.7216 $\pm$ 0.0060	–
Balanced flat RF	0.4773 $\pm$ 0.0168	0.7349 $\pm$ 0.0048	–
Hierarchical RF + Borderline-SMOTE	0.5367 $\pm$ 0.0116	0.7366 $\pm$ 0.0038	0.5110 $\pm$ 0.0148
DLCS	0.5487 $\pm$ 0.0158	0.7389 $\pm$ 0.0034	0.5225 $\pm$ 0.0160
<i>Aggregated LODO over all held-out predictions</i>			
Main, raw	0.3647	0.6649	0.3746
DLCS, raw	0.3680	0.6655	0.3754
Main + smoothing	0.4026	0.6990	0.4051
DLCS + smoothing	0.3993	0.6984	0.4018

The result is not explained by frequent global intervention. Across LODO, the gate makes only 46 raw overrides; 36 are 515→513. After smoothing, room 513 gains 0.0629 F1, while rooms 518 and 511 lose 0.0714 and 0.0476. Four classes (503, 505, 508, and 510) remain at zero F1. Room 505 has only 17 fingerprints and is outside the specialist scope, so this is primarily a coverage limitation.

Figure 2 shows the final LODO confusion matrix. The largest error directions are cafeteria→kitchen, cafeteria→nurse station, kitchen→cafeteria, and hallway→nurse station. The pattern confirms that remaining errors are concentrated in common-area ambiguities and a small number of poorly covered rooms rather than being uniformly distributed.

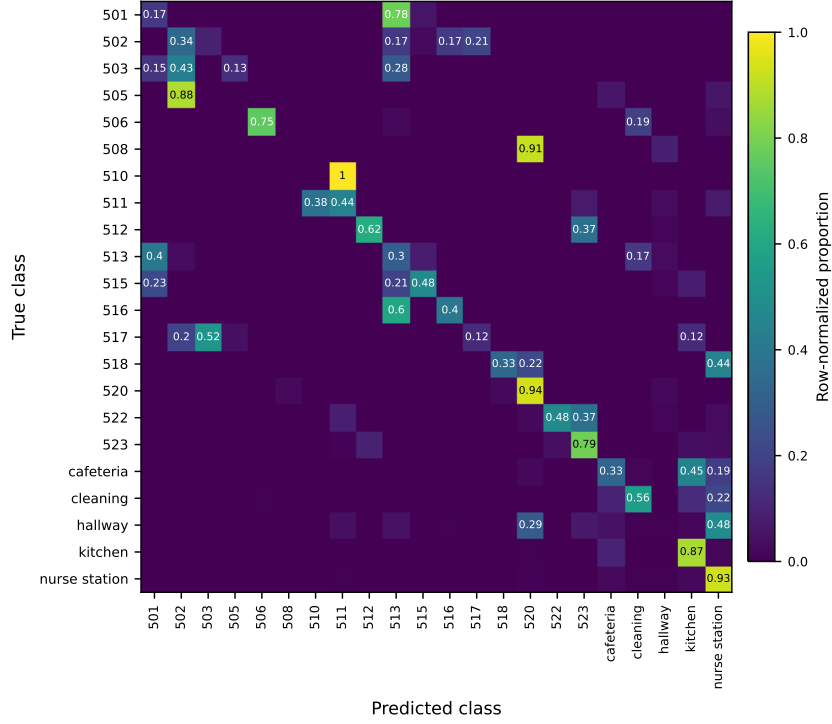


Figure 2. Row-normalized LODO confusion matrix for the final DLCS sequence. Values below 0.12 are omitted for readability.

5.1. Sparse intervention and smoothing interaction

Across LODO, the gate changes only 46 of 15,382 raw predictions (0.30%), so the main model retains 99.70% of decisions. Thirty-six overrides are 515→513. This sparse behavior is central to DLCS: the long-context model is used as a constrained correction layer rather than a global probability blend. The raw aggregate gain is positive, but common temporal smoothing reverses the ordering. This indicates that local correction and sequence post-processing should be designed jointly rather than optimized as independent stages.

Table 4. Smoothed fixed-22-class LODO Macro-F1 by held-out date. The aggregate result in Table 3 is computed over all held-out predictions, not by averaging these four rows.

Held-out date	Main	DLCS	Δ
10 April	0.2337	0.2337	0.0000
11 April	0.3644	0.3644	0.0000
12 April	0.3791	0.3610	-0.0181
13 April	0.3558	0.3707	+0.0149

5.2. Feature dependence and failure cases

Two targeted ablations clarify feature dependence. Removing the 50 main-path count and past-mean-count variables reduces smoothed main LODO Macro-F1 from 0.4026 to 0.3452, so these variables cannot be removed as a cost-free shift safeguard. Removing specialist time-of-day and per-beacon count variables changes final DLCS from 0.3993 to 0.4004, still below the unchanged main model. These variables are retained, but the specialist’s high reliance on time-of-day and the main model’s concentration on beacon 4 should not be interpreted causally. Figure 3 reports model reliance rather than physical causation; impurity importance can favor continuous or frequently split variables.

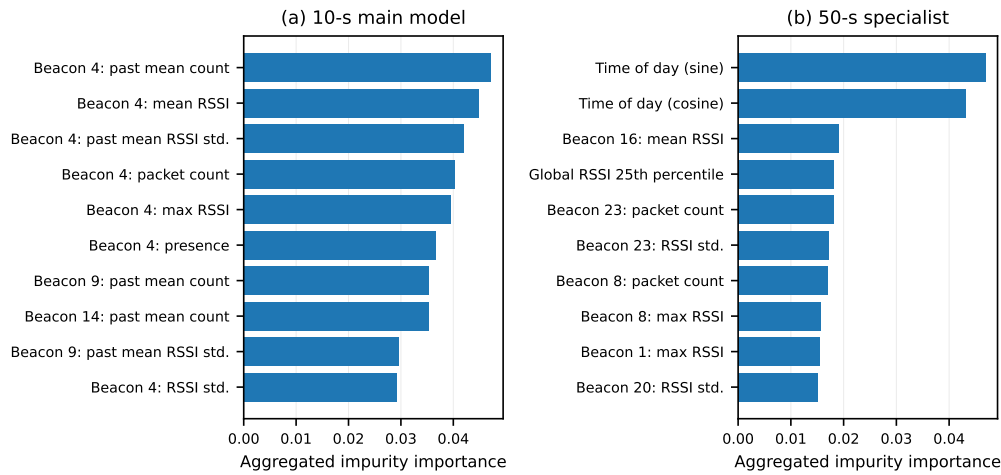


Figure 3. Ten highest model-derived feature importances for the 10-s main path and 50-s specialist.

Four classes (503, 505, 508, and 510) remain at zero F1 in the final LODO sequence. Rooms 505, 516, and 517 are outside the frozen specialist scope; room 505 has only 17 fingerprints and therefore cannot be recovered by the current gate. The main limitations are the four-day sample, classes absent from some training folds, midpoint-labeled transition windows, a facility-specific frozen gate, and non-causal smoothing. Future work should use visit identifiers or LODO during all model selection, learn a gate from out-of-fold meta-features, normalize sensor-rate-sensitive variables, and replace centered smoothing with a causal sequence decoder.

5.3. Evaluation hierarchy and error concentration

The two validation protocols answer different questions. Session-grouped evaluation measures discrimination within the observed deployment period but is optimistic because overlapping windows from one visit can cross folds. LODO removes same-day overlap and tests day separation, although folds holding out an early date can still train on later dates and therefore are not strictly forward-chaining estimates. Both views are retained: the first quantifies the mechanism under the development distribution, and the second bounds its day sensitivity. In the final LODO sequence, the four largest directions—cafeteria→kitchen, cafeteria→nurse station, kitchen→cafeteria, and hallway→nurse station—account for 2,402 of 4,639 errors (51.8%). This concentration suggests that future improvement should target common-area boundaries and day normalization rather than increase specialist coverage indiscriminately.

5.4. Novelty and operational interpretation

DLCS differs from conventional global fusion in three specific ways. The short-window model remains the sole default decision maker; the longer representation is trained only for a restricted patient-room scope; and corrections are directional, pair-constrained, and margin-tested. Consequently, every intervention can be traced to an explicit source class, candidate class, probability margin, and policy condition. This design limits negative transfer and makes the specialist auditable, although the LODO results show that selective correction alone does not solve cross-day signal shift.

5.5. Reproducibility and final inference

The released implementation fixes the feature definitions, class scope, Random Forest settings, correction policy, and temporal filter. Final inference trains only on labeled challenge training data, reconstructs 10-s and 50-s test representations without test labels, maps window predictions back to the original packet order, and appends one location label per organizer row. No test-derived threshold or model selection is used.

6. Conclusion

DLCS combines a hierarchical 10-s main model, a label-free 50-s patient-room specialist, and a directional gate that changes only validated uncertain decisions. It improves Macro-F1 from 0.5367 to 0.5487 under the repeated session-grouped development protocol, but that protocol contains substantial same-visit exposure. Under fixed-22-class LODO, DLCS slightly improves raw predictions (0.3647 to 0.3680), while the final smoothed system scores 0.3993 versus 0.4026 for the main model. The contribution is therefore a selective and interpretable correction mechanism whose within-period benefit is demonstrated, while cross-day improvement remains unresolved.

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References

- [1] C. Garcia and S. Inoue, "Relabeling for indoor localization using stationary beacons in nursing care facilities," *Sensors*, vol. 24, art. 319, 2024, doi: 10.3390/s24020319.

- [2] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, 2009, doi: 10.1109/TKDE.2008.239.
- [3] M. Galar, A. Fernandez, E. Barrenechea, H. Bustince, and F. Herrera, "A review on ensembles for the class imbalance problem: Bagging-, boosting-, and hybrid-based approaches," *IEEE Trans. Syst., Man, Cybern. C*, vol. 42, no. 4, pp. 463–484, 2012, doi: 10.1109/TSMCC.2011.2161285.
- [4] H. Liu, H. Darabi, P. Banerjee, and J. Liu, "Survey of wireless indoor positioning techniques and systems," *IEEE Trans. Syst., Man, Cybern. C*, vol. 37, no. 6, pp. 1067–1080, 2007, doi: 10.1109/TSMCC.2007.905750.
- [5] R. Faragher and R. Harle, "Location fingerprinting with Bluetooth Low Energy beacons," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 11, pp. 2418–2428, 2015, doi: 10.1109/JSAC.2015.2430281.
- [6] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.
- [7] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002, doi: 10.1613/jair.953.
- [8] H. Han, W.-Y. Wang, and B.-H. Mao, "Borderline-SMOTE: A new over-sampling method in imbalanced data sets learning," in *Advances in Intelligent Computing*, LNCS 3644, pp. 878–887, Springer, 2005, doi: 10.1007/11538059_91.
- [9] D. R. Roberts et al., "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017, doi: 10.1111/ecog.02881.
- [10] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Computation*, vol. 3, no. 1, pp. 79–87, 1991, doi: 10.1162/neco.1991.3.1.79.
- [11] F. Zafari, A. Gkelias, and K. K. Leung, "A survey of indoor localization systems and technologies," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2568–2599, 2019, doi: 10.1109/COMST.2019.2911558.